Centre Number	Examination Number

9200200

EXAMINATIONS COUNCIL OF ZAMBIA

Examination for General Certificate of Education Ordinary Level

Design and Technology 6045/1

Paper 1 (Theory & Design)

Friday **8 AUGUST 2025**

Additional Materials
A3 Drawing Paper
Drawing instruments

Time: 2 hours 30 minutes **Marks: 110**

Instructions to Candidates

1 Write the **centre number** and your **examination number** in the spaces provided on this page.
2 Write your **examination number** in the spaces provided on **every page** of this question paper.
3 If working is needed for any question, it must be shown in the spaces provided in the question paper.
4 There are **two** sections in this paper, **A** and **B**. Section B has **three** parts **I**, **II** and **III**.
(i) Answer **all** questions in Section A in the spaces provided in this question paper.
(ii) Section B has **three** parts: Part **I** is compulsory. Write your answers in the spaces provided.
(iii) Choose **one** question from Part **II** and write your answers in the spaces provided.
(iv) Choose **one** question from Part **III** and use the inserted A3 drawing paper provided.
(v) Enter the numbers of questions you have answered from Section B Parts **II** and **III** in the grid provided.
5 Where a freehand sketch is required, it must be drawn to half full size with good linework.
6 At the end of the examination, fasten the drawing paper used securely to the question paper.
7 Use your own discretion where measurements are not provided.

Information for Candidates

1 The number of marks is given in brackets [ ] at the end of each question or part question.
2 Use of colour is allowed.
3 All dimensions are in millimetres.
4 Cell phones and calculators are **not allowed** in the examination room.

For Examiners' Use Only

Question	Mark Obtained	Examiner's Initials
Section A		
Section B		
Part I		
Part II		
Part III		
Total		

©ECZ/GCE/2025/B5

SECTION A: [30 MARKS]

Answer all the questions from this section in the spaces provided in this question paper. If working is needed for any question, it must be shown in the space below that question.

1 (a) **Figure 1** shows two pieces of metal held together by a bolt and nut.

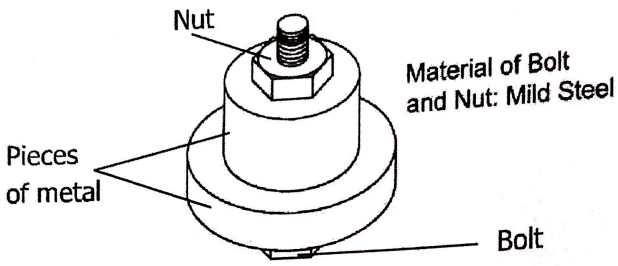


Figure 1

(i) Name a tool you would use to loosen the nut in **Figure 1**.

..... [1]

(ii) State **two** safety precautions to observe when loosening the nut.

..... [1]

..... [1]

(b) **Figure 2** shows a gear train in a Lathe machine.

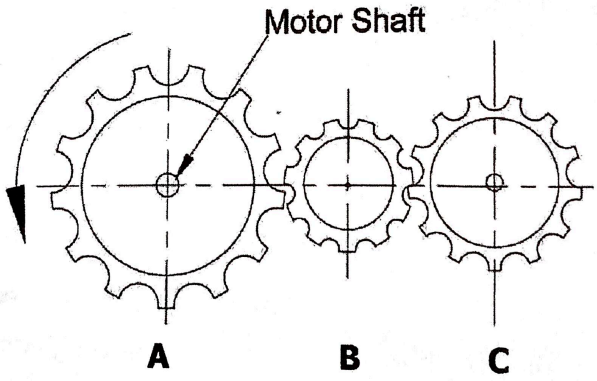


Figure 2

(i) Identify the gear labelled **A.**

..... [1]

(ii) Give the function of the gear identified in **(b)(i).**

..... [1]

(iii) If gear **A** rotates as show in **Figure 2**, indicate on the same diagram the direction in which gear **C** will rotate. [1]

[Total: 6 marks]

Examination Number									

2 **(a)** **Table 1** shows different types of materials. State the appropriate finish you would apply on each.

Table 1

Material	**Finish**	
Stainless		[1]
Acrylic		[1]
Block Board		[1]

(b) **Figure 3** shows a mechanism used to transmit motion.

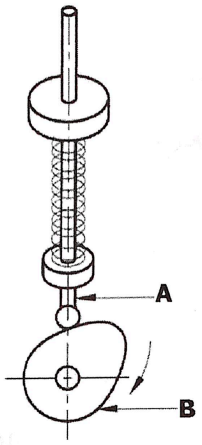


Figure 3

(i) Identify part **A** and **B**.

A ..... [1]

B ..... [1]

(ii) Mention a device where the mechanism in **Figure 3** can be applied.

..... [1]

[Total: 6 marks]

3 **(a)** **Figure 4** shows **two** different cutting tools.



A **B**

Figure 4

(i) State the function of;

Tool **A** ..... [1]

Tool **B** ..... [1]

Examination Number

(ii) Mention a specific material for making cutting edges of both tools.

..... [1]

(b) Provide the correct term for each of the following definitions:

(i) The resource in form of cash or goods needed to start a business,

..... [1]

(ii) A person who runs a business/company on his own or as an individual.

..... [1]

(c) **Figure 5** shows a shape of a plain figure.

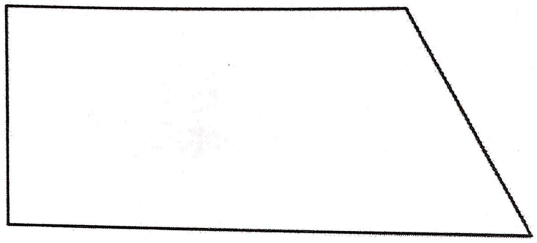


Figure 5

Identify the shape shown.

..... [1]

[Total: 6 marks]

4 **Figure 6** shows a tool used in a Design and Technology workshop.

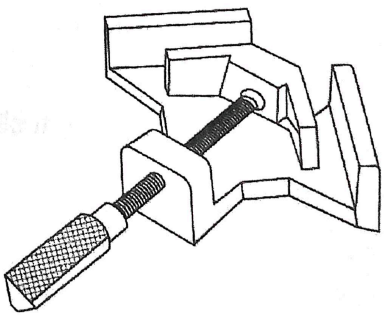


Figure 6

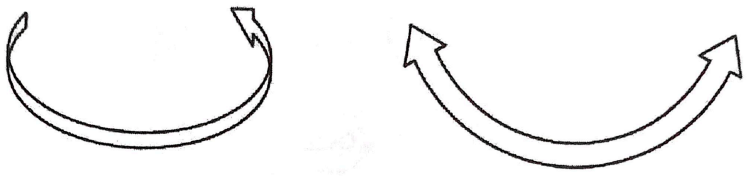
(a) **(i)** Identify the tool.

..... [1]

(ii) State the function of the tool identified in **a(i)**.

..... [1]

(b) **Figure 7** shows a symbol used for motion.



A **B**

Figure 7

Mention **one** device that uses:

(i) Motion **A** ..... [1]

(ii) Motion **B** ..... [1]

(c) State any **two** attributes of an entrepreneur.

..... [1]

..... [1]

[Total: 6 marks]

5 **Figure 8** shows a sign for a clinic.

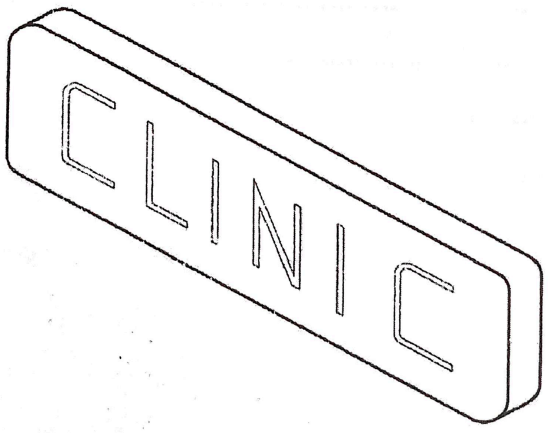


Figure 8

(a) **(i)** Suggest a specific type of material that can be used to make the sign.

..... [1]

(ii) Mention **one** property of the material suggested in **a(i).**

..... [1]

Examination Number

(b) **Figure 9** shows part of a grinding machine used to sharpen chisels.

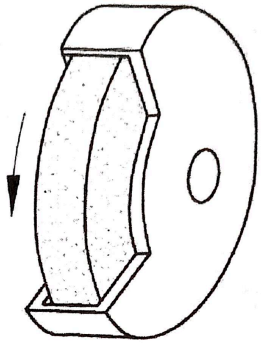


Figure 9

Mention **two** attachments that are fixed to the grinding machine for the safety of users.

(i) ..... [1]

(ii) ..... [1]

(c) SWOT Analysis is one of the methods used to assess business ideas.

State **two** advantages of using SWOT Analysis.

(i) .....

.....

..... [1]

(ii) .....

.....

..... [1]

[Total: 6 marks]

Examination Number									

SECTION B: [80 MARKS]

There are three parts in this section; Part I, II and III. Part I is compulsory. Answer one question from Part II and one question from part III. Write your answers in the spaces provided.

Part I: Compulsory

6 Study **Figure 10** and answer the questions that follow.

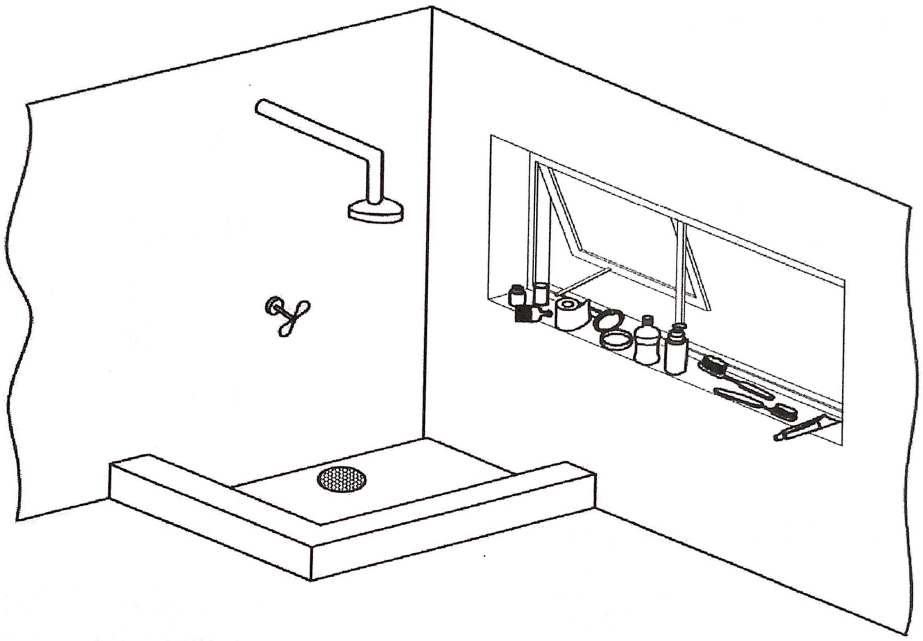


Figure 10

(a) (i) Write the situation obtaining in **Figure 10**.

.....

..... [2]

(ii) Identify the **problem** from the situation in **(a)(i)**.

.....

..... [2]

(iii) Formulate a design brief to address the problem identified in **(a)(ii)**.

.....

..... [2]

Examination Number									

(b) **(i)** Using annotations, sketch in pictorial view, one possible solution to the problem identified in **(a)(ii)**

[10]

(ii) Suggest **two** specific materials you would use for your solution in **b(i).**

..... [1]

..... [1]

(iii) Suggest **four** important aspects you would consider when developing chosen solution.

..... [1]

..... [1]

..... [1]

..... [1]

(c) **(i)** Give **two** aspects you would consider when evaluating a product.

..... [1]

..... [1]

(ii) Mention **one** safety precaution you would consider during reaction of a possible solution.

..... [1]

[Total: 25 marks]

Part II

Answer one question from this part in the spaces provided.

7 **(a)** **Figure 11** shows a simple electronic switching circuit.

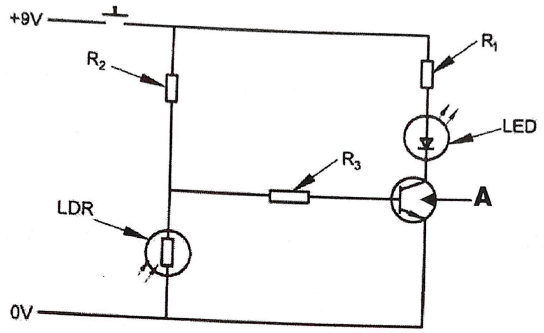


Figure 11

(i) Identify the component labelled **A**.

..... [1]

(ii) Explain the function of the following components in the circuit.

A LDR

..... [2]

B LED

..... [2]

(iii) Give **one** reason why resistors are important in electronic circuits.

.....

..... [2]

(b) **Figure 12** shows two metallic square tubes to be joined using an adhesive.

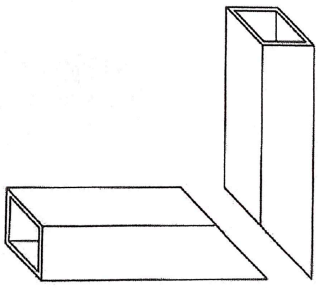


Figure 12

(i) Suggest a suitable adhesive that can be used to join the metal tubes.

..... [1]

(ii) Explain **one** safety precaution to be observed when using the adhesive.

.....

..... [2]

Examination Number									

(c) **Figure 13** shows a juice bottle with labels.



Figure 13

(i) Suggest a suitable material for making the bottle.

..... [2]

(ii) Mention a property of material suggested in **(c)(i)** which makes it suitable for packaging drinks.

..... [2]

(iii) Give a reason why labelling of products is important.

.....

..... [2]

(d) **Figure 14** shows a solar water heater.

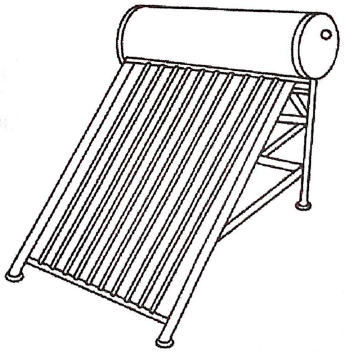


Figure 14

(i) What are the **two** advantages of using solar power to heat water?

.....

..... [2]

(ii) Give **two** disadvantages of using solar power.

.....

..... [2]

Examination Number									

(e) **Figure 15** shows a hook which is part of the tool rack made in a workshop.

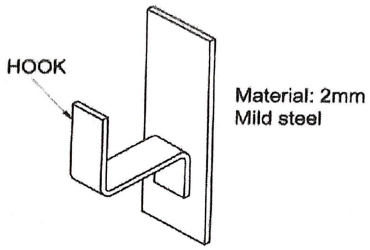


Figure 15

(i) Explain why the hook should be annealed before bending it.

.....

..... [2]

(ii) Describe how to anneal the hook.

.....

.....

..... [3]

[Total: 25 marks]

8 **(a)** **Figure 16** shows a door fitted to the frame.

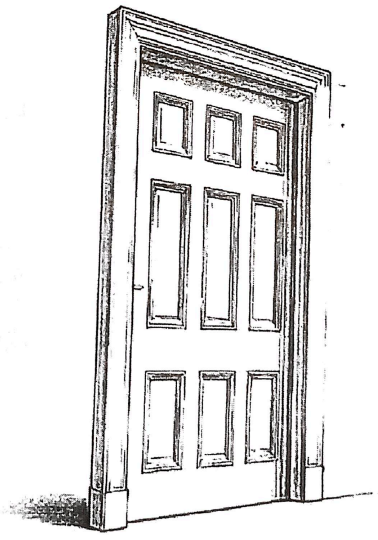


Figure 16

(i) Mention a suitable fixing device you would use to fit the door to the frame.

..... [1]

Examination Number									

(ii) Give **one** reason for using the fixing device mentioned in **(a)(i)**.

.....

..... [2]

(iii) With the help of sketches, show how to fix the door to the frame.

[4]

(b) **Table 2** shows items and materials from which each item would be made.

Table 2

Item	**Material**
Cooking pot	Aluminium
TV and Radio packages	Expanded Polystyrene
Chairs and Tables	Plywood.

Why is the material shown for each item suitable for use in each case?

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

(c) It is important to carryout Market Research of products and services when starting a business.

Give **two** reasons why Market Research is important.

.....

.....

.....

..... [4]

(d) **Figure 17** shows a Ballpein hammer.

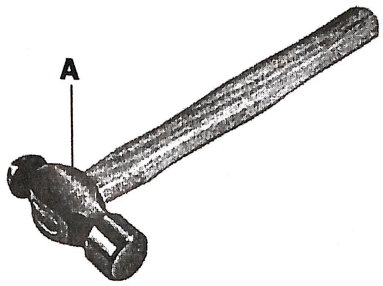


Figure 17

(i) Suggest a specific material for making part **A.**

..... [1]

(ii) Give a reason for using the material in **(d)(i)** to make part **A.**

.....

.....

..... [2]

(iii) Explain how you would take care of a hammer in a workshop.

.....

.....

.....

..... [2]

Examination Number									

(e) Sketch a sliding bevel in pictorial view.

[3]

[Total: 25 marks]

9 **(a)** **Figure 18** shows a circuit diagram.

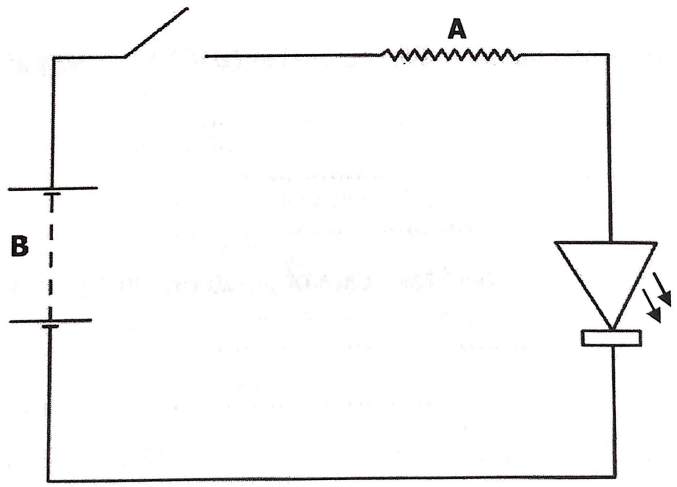


Figure 18

(i) State the functions of components **A** and **B**.

.....

.....

.....

..... [4]

(ii) Give **two** types of circuit boards on which the circuit diagram can be mounted.

.....

.....

[2]

(b) **Figure 19** shows tools used in a workshop.



Figure 19

(i) Identify the tools **X** and **Y** in **Figure 19**.

X ..... [1]

Y ..... [1]

(ii) Using a sketch, show how tool **Y** is used.

[3]

(iii) Explain **one** way of maintaining the tools shown in **Figure 19**.

.....

..... [1]

Examination Number

(c) **Figure 20** shows a rolling pin made in a workshop.

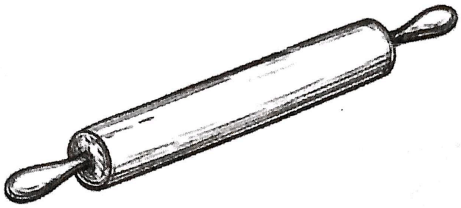


Figure 20

(i) Mention **one** machine that can be used to make a rolling pin

..... [1]

(ii) Suggest **two** safety precautions you would observe when making the rolling pin on the machine mentioned in **(c)(i)**.

.....

..... [2]

(iii) Describe **one** way of caring and maintaining the machine stated in **(c)(i)**.

.....

..... [1]

(d) **Figure 21** shows an exploded view of a garden rake.

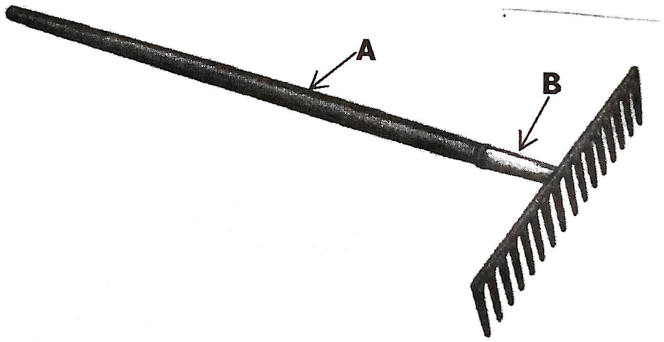


Figure 21

(i) Mention a specific material that can be used to make parts **A** and **B**.

A ..... [1]

B ..... [1]

(ii) What suitable method would you use to join parts **A** and **B**?

.....

..... [1]

(iii) Using annotated sketches, show how to join part **A** to **B**.

[4]

(iv) Explain the importance of applying a finish to a garden rake.

.....

.....

.....

..... [2]

[Total: 25 marks]

Examination Number

Part III

Answer one question on the inserted A3 drawing paper provided.

10 A design of a bridge is to be constructed across a stream. The bridge is made from concrete and steel. A front Elevation of a bridge is shown in **Figure 22.**

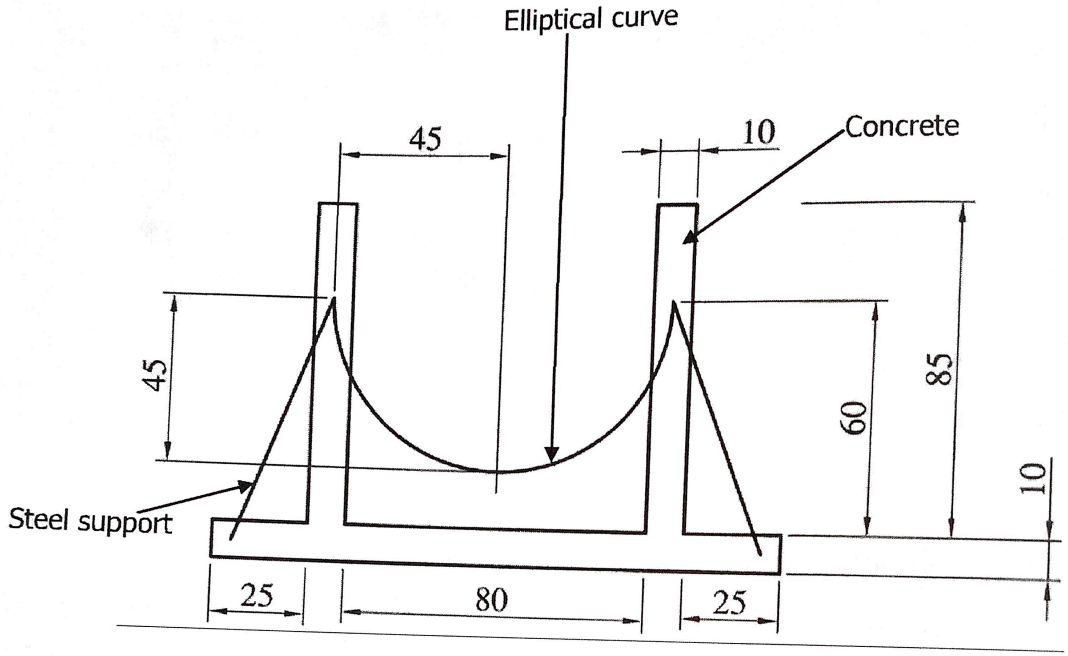


Figure 22

(a) Construct the bridge shown. [20]

(b) Design a suitable sign post to caution motorists of the bridge ahead. [7]

(c) Render your signpost using appropriate techniques. [3]

[Total: 30 marks]

11 A design of a funnel to be used in the science laboratory is to be made. The funnel is made from hollow plastic material. A front Elevation of the funnel is shown **Figure 23**.

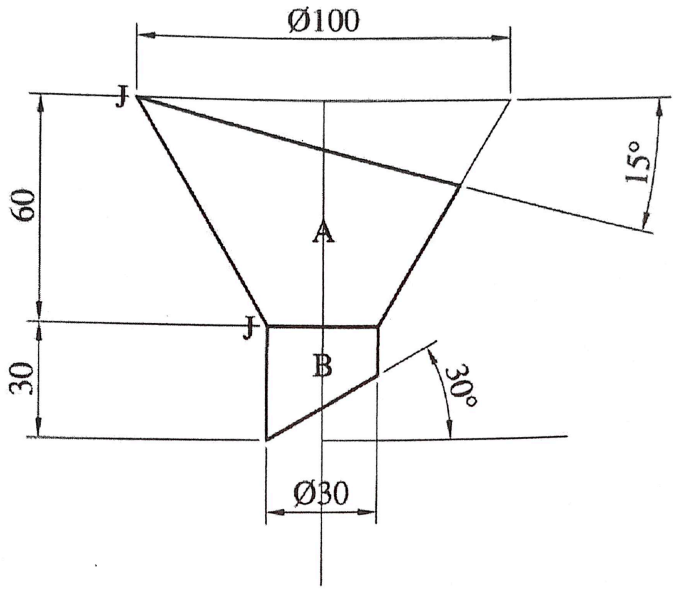


Figure 23

(a) Copy the given view. [7]

(b) Construct the surface development of part "A" of the funnel. The opening is at **J-J**. [15]

(c) In a regular hexagon with distance A/F = 90, design a sign to illustrate graphically the information "Handle with care." [8]

[Total: 30 marks]